Geographic Birthplace as a Predictor of Primary Language:

A Cross-National Observational Study

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Abstract

Background: Language acquisition is a fundamental aspect of human development, yet the relative contributions of environmental versus biological factors remain underexplored in large-scale empirical studies. This observational study examines whether geographic birthplace predicts primary language spoken across multiple nations.

Methods: We analysed national census data from eight countries (N = 1,811,487,320 individuals) spanning six continents. Primary outcome was concordance between country of residence and dominant national language spoken. Chi-square tests and effect size calculations (Cohen's h) were conducted.

Results: Across all nations examined, geographic residence demonstrated strong concordance with national language acquisition (range: 72.0%–96.9%). Effect sizes ranged from $h = 0.46$ to $h = 1.22$ (mean = 0.93; classified as "medium" to "large" by conventional standards). The observed pattern held regardless of the specific language examined.

Conclusions: Geographic environment appears to be an extraordinarily strong predictor of language acquisition. Supplementary analysis of international adoption studies and twin research confirms that environment, not genetics, determines which language an individual speaks. The implications for understanding human behavioural acquisition more broadly warrant further investigation.

Keywords: language acquisition; environmental factors; cross-national study; census data; nature versus nurture

Key Points

- Geographic environment predicts primary language with 72–97% accuracy across eight nations
- Effect sizes (mean Cohen's $h = 0.93$) exceed conventional "large" thresholds
- International adoption studies show 100% language replacement regardless of genetic ancestry
- Twin studies confirm genetics affects language *ability*, not *which* language is spoken
- Findings have implications for understanding environmental determination of complex human behaviours

1 Introduction

1.1 Background

Human language is among the most complex cognitive abilities exhibited by any species. The average adult possesses a productive vocabulary of approximately 20,000–35,000 words, applies grammatical rules unconsciously in real-time, and processes speech at rates exceeding 150 words per minute (Brysbaert et al., 2016). Despite this complexity, healthy children across all cultures acquire language with remarkable consistency.

The question of how language is acquired has been debated extensively. Nativist perspectives emphasise innate language acquisition devices (Chomsky, 1965), while empiricist perspectives highlight environmental exposure and social learning (Tomasello, 2003). However, large-scale empirical studies examining the actual distribution of language outcomes across populations remain limited.

1.2 Research Question

This study addresses a straightforward empirical question: *To what extent does geographic birthplace predict the primary language an individual speaks?*

We then performed a systematic cross-national analyses quantifying this relationship with standardised effect size metrics.

1.3 Hypotheses

H_0 (Null Hypothesis): Geographic birthplace is not associated with primary language spoken, and language acquisition occurs independently of geographic environment.

H_1 (Alternative Hypothesis): Geographic birthplace is associated with primary language spoken, and language acquisition is related to geographic environment.

We set our significance threshold at $\alpha = 0.05$.

2 Methods

2.1 Design

Cross-sectional observational study using publicly available national census data.

2.2 Data Sources

We identified national statistical agencies with publicly available census data on language spoken. Countries were selected based on: (1) availability of recent census data (2011–2022); (2) data published in or translatable to English; (3) inclusion of language variables; and (4) geographic and linguistic diversity.

2.3 Variables

- **Predictor**: Country of residence at time of census
- **Outcome**: Primary language spoken (national language vs. other)

2.4 Statistical Analysis

For each country, we calculated:

1. Proportion speaking the dominant national language
2. Chi-square goodness-of-fit test against null expectation (50%)
3. Effect size (Cohen's h)

Cohen's h is calculated as:

$$h = 2\arcsin(\sqrt{p_1}) - 2\arcsin(\sqrt{p_2}) \tag{1}$$

Effect size interpretation (Cohen, 1988): Small = 0.20; Medium = 0.50; Large = 0.80.

3 Results

3.1 Sample Characteristics

The combined sample included N = 1,811,487,320 individuals from eight countries (Table 1).

Table 1: Sample Characteristics by Country

Country	Year	Population	Source
Australia	2021	25,422,788	ABS
Canada	2021	36,991,981	Statistics Canada
China	2020	1,411,778,724	NBS
France	2021	67,390,000	INSEE
Germany	2022	82,700,000	Destatis
Mexico	2020	126,014,024	INEGI
New Zealand	2018	4,699,755	Stats NZ
United Kingdom	2021	56,490,048	ONS
Total	—	**1,811,487,320**	—

3.2 Language Concordance

Table 2 presents the proportion speaking the dominant national language.

Table 2: Proportion Speaking Dominant National Language

Country	Language(s)	% Speaking
Australia	English	72.0%
Canada	English/French	96.9%
China	Chinese	92.0%
France	French	91.2%
Germany	German	81.0%
Mexico	Spanish	93.8%
New Zealand	English	95.4%
United Kingdom	English	91.1%

3.3 Statistical Tests

All chi-square tests were significant at $p < .001$ (Table 3).

3.4 Effect Size Summary

Mean Cohen's h = **0.93** (SD = 0.24), range 0.46–1.22.

This exceeds the conventional "large" threshold of 0.80 by a factor of 1.2.

Table 3: Chi-Square and Effect Size Results

Country	%	χ^2	df	p	Cohen's h
Australia	72.0	4,921,852	1	$< .001$	0.46 (S)
Canada	96.9	32,547,173	1	$< .001$	1.22 (L)
China	92.0	996,151,068	1	$< .001$	1.00 (L)
France	91.2	45,756,193	1	$< .001$	0.97 (L)
Germany	81.0	31,789,880	1	$< .001$	0.67 (M)
Mexico	93.8	96,700,138	1	$< .001$	1.07 (L)
New Zealand	95.4	3,874,779	1	$< .001$	1.14 (L)
United Kingdom	91.1	38,169,422	1	$< .001$	0.96 (L)

Note: S = Small, M = Medium, L = Large effect size.

4 Discussion

4.1 Summary

Across eight nations representing 1.8 billion individuals:

1. Proportions speaking national language: 72.0%–96.9%
2. All p-values $< .001$
3. Mean effect size exceeds "large" threshold
4. Pattern consistent across languages and regions

4.2 Supplementary Evidence

4.2.1 International Adoption Studies

Korean children adopted by Swedish families speak Swedish. Chinese children adopted by American families speak English. No study has found spontaneous birth-language acquisition without environmental exposure (Pallier et al., 2003; Hyltenstam et al., 2009).

Effect: 100% language replacement.

4.2.2 Twin Studies

The Minnesota Study (Bouchard et al., 1990) found that genetics affects language *ability* (heritability 25–70%), but *which language* is spoken shows 0% heritability.

The "Jim Twins" spoke English because both were raised in Ohio—not due to genetic predisposition.

4.2.3 Generational Studies

Hispanic immigrants show complete language shift within three generations (Portes & Rumbaut, 2001):

- 1st generation: 85% Spanish-dominant
- 3rd generation: 92% English-dominant

4.3 Implications

The finding that "people speak the language of their environment" is obvious. Yet we apply different logic to other behaviours. Our entire criminal justice system is predicated on it.

A child in Sydney speaks English because of environment. Might a child raised where anger is standard also learn anger from environment?

> *If the most complex behaviour (language) is 100% predicted by environment, the default assumption for simpler behaviours might reasonably be environmental as well.*

5 Conclusions

Geographic environment strongly predicts language acquisition (mean $h = 0.93$, all $p < .001$). This holds across eight nations, multiple languages, and 1.8 billion individuals.

The magnitude of environmental determination of language—one of the most complex human behaviours—may have implications for understanding behavioural acquisition more broadly.

References

Bouchard, T. J., Lykken, D. T., McGue, M., Segal, N. L., & Tellegen, A. (1990). Sources of human psychological differences: The Minnesota Study of Twins Reared Apart. *Science*, 250(4978), 223–228.

Brysbaert, M., Stevens, M., Mandera, P., & Keuleers, E. (2016). How many words do we know? *Frontiers in Psychology*, 7, 1116.

Chomsky, N. (1965). *Aspects of the theory of syntax.* MIT Press.

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum.

Hyltenstam, K., et al. (2009). Dominant-language replacement: The case of international adoptees. *Bilingualism: Language and Cognition*, 12(2), 121–140.

Pallier, C., et al. (2003). Brain imaging of language plasticity in adopted adults. *Cerebral Cortex*, 13(2), 155–161.

Portes, A., & Rumbaut, R. G. (2001). *Legacies: The story of the immigrant second generation.* UC Press.

Tomasello, M. (2003). *Constructing a language.* Harvard University Press.